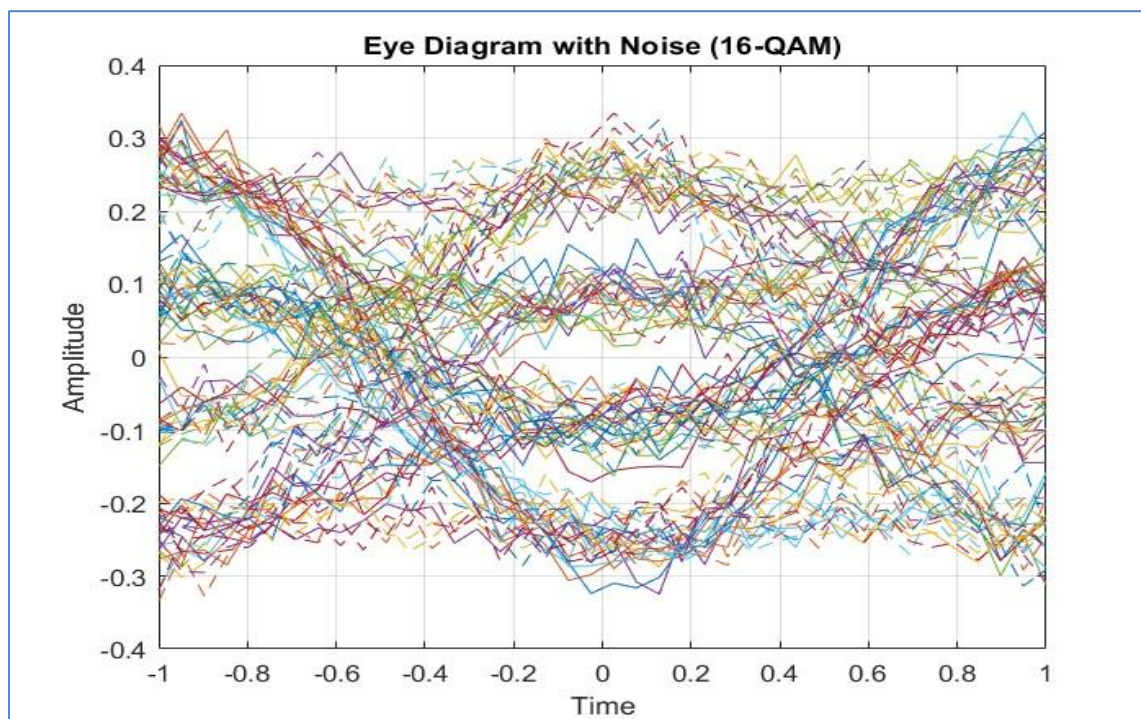
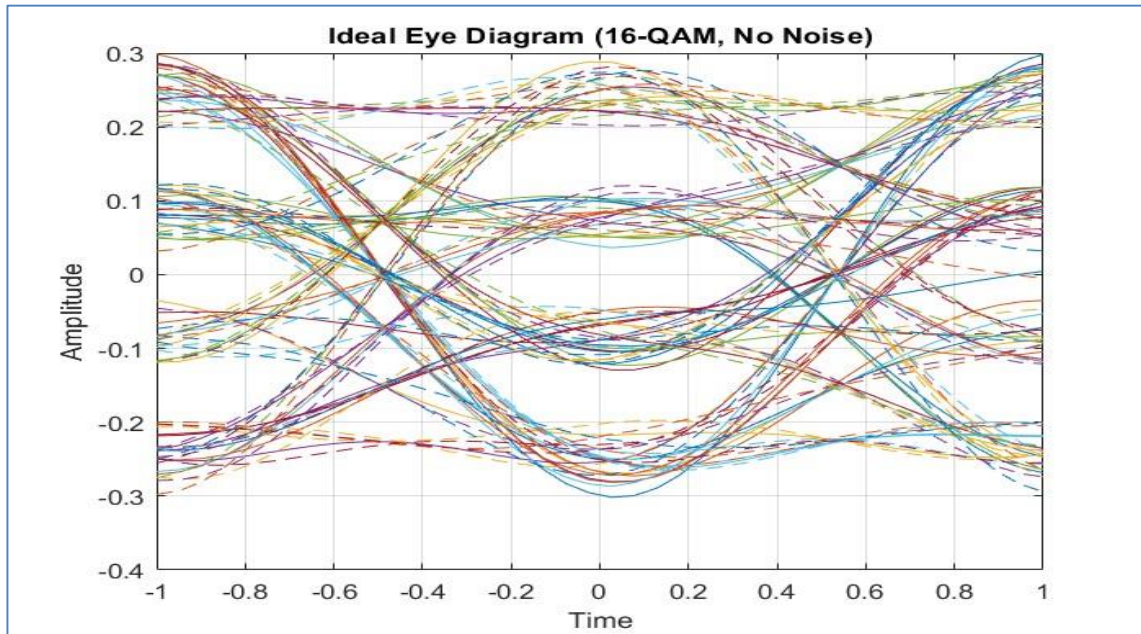


Simulate a Basic Digital Communications Link- Assignment -4-

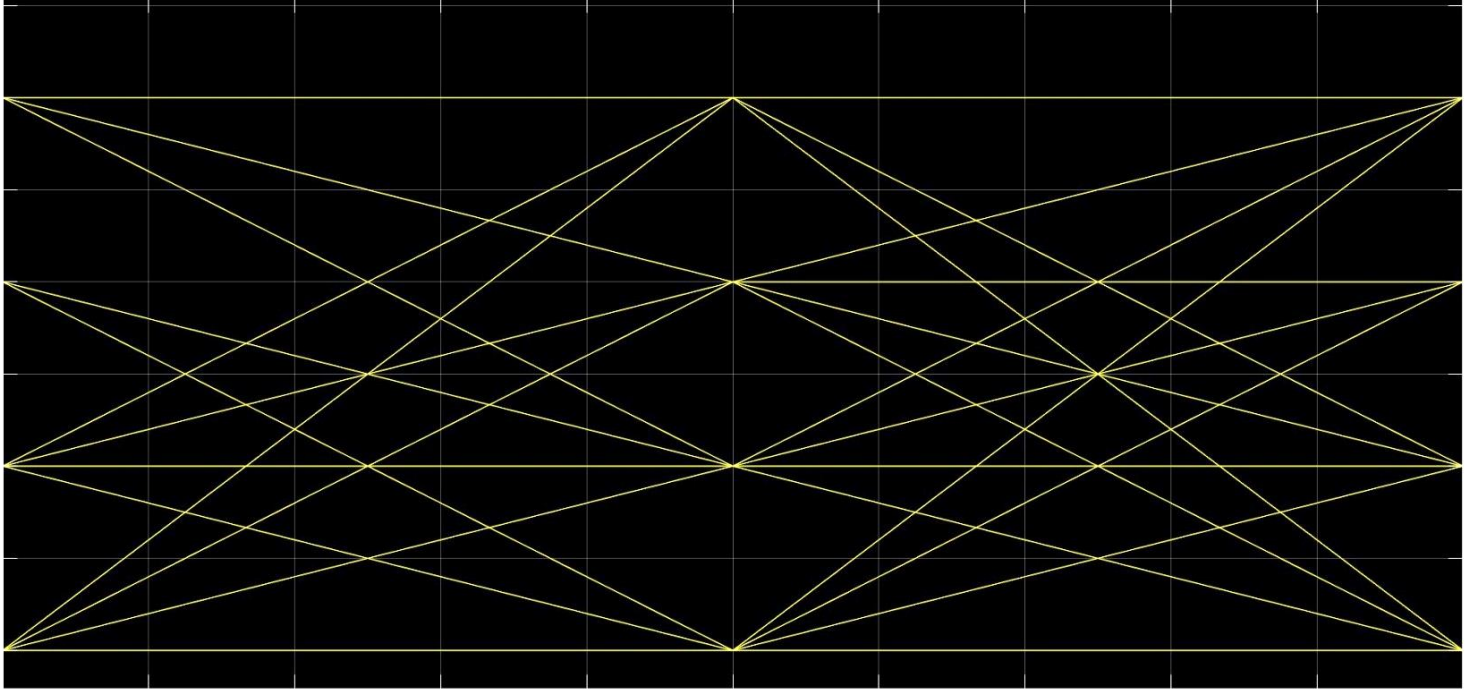
Assignment: Implement the MATLAB code as in following script file, should be named MATLAB file as (HW4.m), Make sure to follow the following requirements:

Note: Re-write the MATLAB scripts based on our course rules, make cells, cell title, and in line descripts and units as discussed in our lecture.

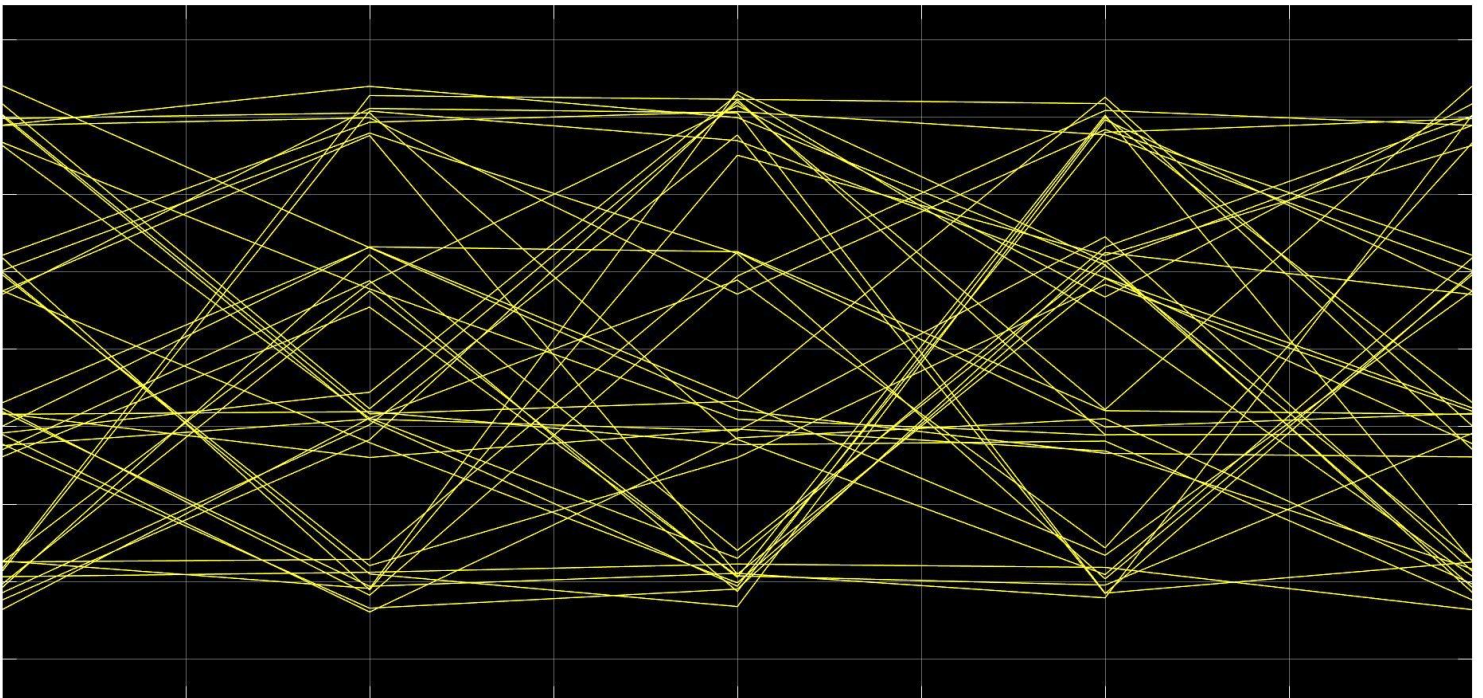
Part A: By using Matlab script, model a communication link (TX, QAM modulation, channel with noise, and RX), then generate an eye diagram to analyze the recovered signals and study the effect of noise as the following figures. **(50 points)**



Part B: By using Matlab Simulink, model a communication link (TX, QAM modulation, channel with noise, and RX), then generate an eye diagram to analyze the recovered signals and study the effect of noise as the following figures. **(50 points)**



Ideal Eye Diagram (16-QAM, No Noise)



Eye Diagram with Noise (16-QAM)